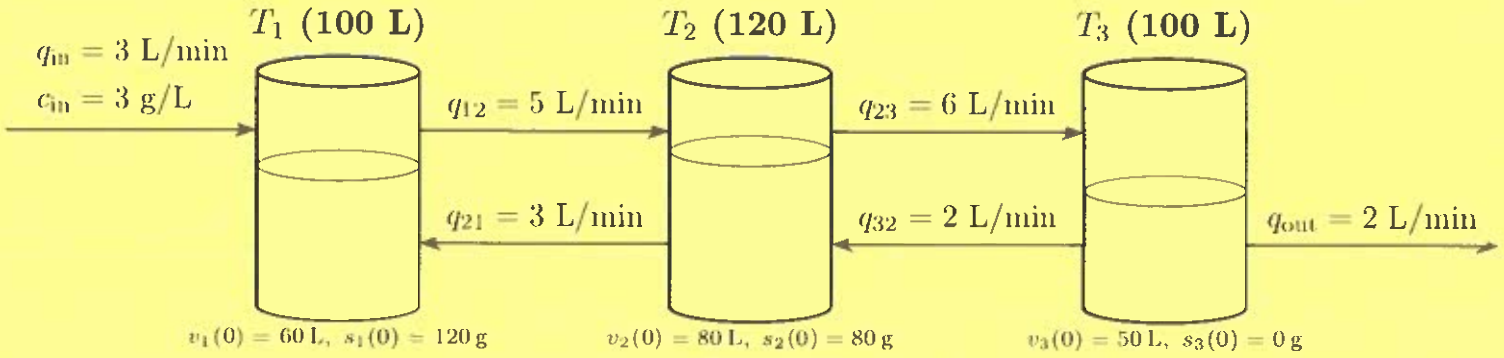
MA3521 — Well-Mixed Tanks

Name: Key

Problem 1

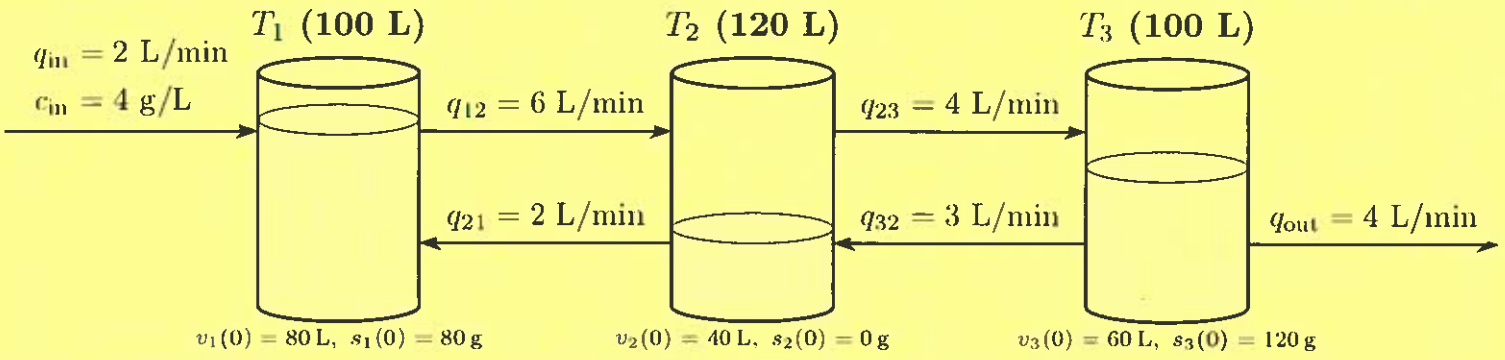


The IVP $\mathbf{s}' = A\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \mathbf{s}_0$ makes sense for $0 < t <$ 25 with

$$\mathbf{v}(t) = \begin{pmatrix} 60+t \\ 80-2t \\ 50+2t \end{pmatrix} \quad \mathbf{s}_0 = \begin{pmatrix} 120 \\ 80 \\ 0 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 9 \\ 0 \\ 0 \end{pmatrix}$$

$$A = \begin{pmatrix} -5/V_1 & 3/V_2 & 0 \\ 5/V_1 & -9/V_2 & 2/V_3 \\ 0 & +6/V_2 & -4/V_3 \end{pmatrix}$$

Problem 2



The IVP $\mathbf{s}' = A\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \mathbf{s}_0$ makes sense for $0 < t <$ 20 with

$$\mathbf{v}(t) = \begin{pmatrix} 80 - 2t \\ 40 + 3t \\ 60 - 3t \end{pmatrix} \quad \mathbf{s}_0 = \begin{pmatrix} 80 \\ 0 \\ 120 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 8 \\ 0 \\ 0 \end{pmatrix}$$

$$A = \begin{pmatrix} -6/v_1 & +2/v_2 & 0 \\ 6/v_1 & -6/v_2 & 3/v_3 \\ 0 & 4/v_2 & -7/v_3 \end{pmatrix}$$